

Motion Trajectory Reconstruction

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Abstract

This paper investigates the feasibility of decoding continuous movement from EEG using deep learning. We implement and compare three distinct architectures: a hybrid Convolutional Neural Network and Long Short-Term Memory (CNN+LSTM) model, a modified Pix2Pix Generative Adversarial Network (GAN) adapted for signal-to-signal translation, and a Convolutional Autoencoder (CAE). These models are evaluated on two publicly available datasets representing different levels of motor complexity: a constrained grasp-and-lift task (WAY-EEG-GAL) and unconstrained full-body locomotion. We use the Pearson Correlation Coefficient (PCC) to measure decoding accuracy. Our results show that all models can successfully reconstruct motion trajectories in the constrained task, with the GAN architecture achieving slightly superior performance (average PCC > 0.70). However, performance degrades significantly for the unconstrained locomotion task (average PCC < 0.45), highlighting the challenges posed by motion artifacts and increased degrees of freedom. This work underscores the potential of deep learning for motion trajectory reconstruction while also outlining the key obstacles that must be overcome for real-world clinical applications.

1 Introduction

Electroencephalogram (EEG)-based Brain-computer interfaces (BCIs) have shown great promises in providing an alternative communication or control method for individuals with motor disabilities by bridging human thoughts with external devices. BCI can be achieved with invasive or non-invasive technologies. Invasive technologies such as the **ECOG** provide a better signal-to-ratio and spatial resolution of the signals. However, they require complex implantation surgery and long term maintenance. non-invasive **electroencephalography (EEG)**-based BCI systems are safer, cheaper, and offer high temporal resolution but suffer from signal artifacts due to volume conduction. Most of the research and datasets available focus on the Motor Imagery domain of BCI systems. **Motor imagery (MI)** is defined as a mental process where an individual rehearses or simulates a given action without actually executing it. The works on Motor Imagery result in discrete classification models which are insufficient for the continuous control of exo-skeletons or prosthetics. Therefore, we chose to focus on Motor Execution or **Motion**

Trajectory Reconstruction. During the execution of the task, neural signals (EEG signals) and dynamic data (motion trajectories) are recorded in parallel. Being able to decode this continuous movement would allow for smoother control of the devices, which is a requirement for complex activities.

2 Related Works

Research in Motor Trajectory Reconstruction (MTR) shows considerable progress. In the early works of MTR, the most commonly used motion trajectory reconstruction algorithms were Kalman filter, and Multivariate Linear Regression (MLR) Bradberry et al. [1]. The models used EEG voltage, band-power features, or just raw EEG amplitudes. With the success and popularity of Deep Learning, it started being used in MTR. With DL it was less important to extract EEG features manually and more of a unified decoding framework. Convolutional Neural Networks (CNN's) were used since they are good at spatial data extraction, and later Long Short Term Memory (LSTM's) were used to capture temporal dependencies in the work of X. Wang et al. [2]. Combining CNN and LSTM became a common architecture. Recently, the Transformer model was used on a dataset of hand motion data of executing Chinese sign language sentences in the research of P. Wang. et al. [3]. The method achieved the highest results yet, suggesting that it can produce better feature representations in the spatial dimension when compared to other deep learning methods.

A problem found across all of this research is the lack of dataset availability, which means no possibility to verify their results. Nevertheless, We did find several papers co-authored by "Anant Jain" and "Lalan Kumar" et al. which used a publicly available dataset "**WAY-EEG-GAL**" which comprises of healthy participants grasping and lifting an object for 3,936 trials in total. And as all research in MTR does, they used the **Pearson Correlation Coefficient (PCC)** to measure the similarity between the predicted motions and the real motions. They achieved a high score, so we wanted to recreate their methods to verify the results. Pancholi et al. [4] The first paper achieves the best results by using the hybrid architecture of **CNN+LSTM**. They preprocess the EEG data by using Band-pass Filtering, Common Average Referencing, Independent Component Analysis for artifact removal, and Channel Selection.

The band-pass filtering was done to obtain different frequency ranges of Delta(0.5-3Hz), Theta(3-7Hz), Alpha(7-12Hz), and All (0.5-12Hz). All (0.5-12Hz) resulted in the best PCC scores. They applied a method known as **Standardized Low Resolution Brain Electromagnetic Tomography (sLORETTA)** to visualize the activations in the brain. sLORETTA takes in the 2D scalp surface EEG data into 3D representation of the entire cortex. They used this to determine the time at which the motor cortex is most activated to decide on the time lag to use for the EEG segments. It was concluded that the EEG source localization can provide the intended hand movement information approximately 350ms prior to the actual hand movement. The average of all subjects' PCC results achieved in this paper are **0.86, 0.90, 0.80** for the **X Y Z** axes. In the next paper Anant Jain et al. [5] "PreMovNet" they use only the Delta band (0.5-3Hz) with a 250ms time lag and achieve slightly lower PCC average scores 0.7908, 0.7990 and 0.6005. They published a few papers after this around the same topic and dataset but the results didn't improve. The problem with these research papers is that they left out important specifics and details such as:

- Exactly which part of the trials did they use for training?
- What was the size of the EEG segment used?
- If the model predicted only a single point for each EEG segment, then what is the step-size between each segment?
- Which position point of the hand were they predicting?

3 Foundations

3.1 Datasets

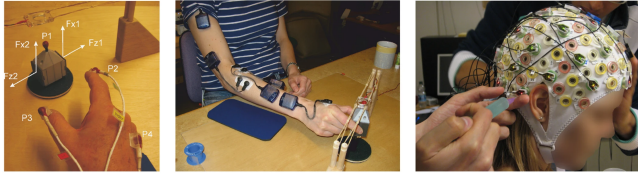


Figure 1. WAY-EEG-GAL dataset

3.1.1 First dataset "WAY-EEG-GAL". In this dataset, 12 participants performed a grasp and lift task of a small object for a total of 3,936 trials. "For each trial the participant was cued to reach for the object, grasp it with the thumb and index finger, lift it and hold it for a couple of seconds, put it back on the support surface, release it, and, lastly, to return the hand to a designated rest position. The signals that were recorded are EEG (32 channels), EMG (five arm and hand muscles), the 3D position of both the hand and object. We are only interested in the EEG, and 3D position of the hand. We used the X,Y,Z positions of the wrist point for all of the training, and decoding.



Figure 2. Experimental gait course "Full body in unconstrained motion"

3.1.2 Second dataset "Full body in unconstrained motion". A 64-channel Ag/AgCl active electrode EEG setup was used to wirelessly record from the face and scalp (sampling frequency=1000Hz) of ten healthy individuals (five males, five females; ages 18-31 years old)

"The subjects were instrumented with 17 wireless inertial measurement units (IMU's) for full body motion capture at 30Hz, and 60Hz for some trials. The experiments were conducted on a custom-built gait course (shown in Fig. 4) incorporating five steady locomotion modes: level ground walking (LW), stair descent (SD), stair ascent (SA), ramp descent (RD), and ramp ascent (RA)"

3.2 Pre-processing

3.2.1 "WAY-EEG-GAL". Data Importing: The continuous kinematic and EEG data are available in (.mat) files with the following format HS_P(Participant_number)_S(Session_number). There are 9 sessions for each participant, and each session contains from 29 to 34 trials. Our data importing script loops through all the files and extracts X, Y, and Z axes position data of the hand. As well as the EEG data. Then, the script saves the data in respective .npy files so we don't have to extract data from the .mat files each time.

Data Preprocessing: The EEG and the Kinematic data had the same sampling frequency of 500 Hz so I didn't need to downsample any signals. The only preprocessing done on the kinematic data was the windowing and normalization in the training script. The EEG signals needed more preprocessing. There are many available tools for EEG data processing, such as "Fieldtrip", "MNE", and the most popular tool "EEGLAB" which is a toolbox in MATLAB. We used the MNE python library for all the EEG signal processing due to its easy integration, good documentation, and availability of all needed functions and algorithms. The preprocessing steps we followed were:

- Bandpass Filtering
- Common Average Re-referencing
- Independent Component Analysis for artifact removal
- Channel Selection

We performed **Bandpass Filtering** on the data. We tried different ranges of (0.5-3Hz), (0.5,12Hz), and (0.5-40Hz). The

(0.5-40Hz) range gave the best results.

Common Average Re-referencing is achieved by creating an average of all scalp channels and subtracting the resulting signal from each channel. Some papers show it reduces noise.

Independent Component Analysis (ICA) is a signal processing method to separate independent sources linearly mixed in several sensors. In EEG it is used to separate the artifacts from the clean data. The artifacts can be eye movements, blinks, or EMG artifacts from the muscles. Unfortunately to take full advantage of the components produced by the algorithm, an expert is needed, but there are alternative methods to identifying the artifacts. We used 2 methods to detect if the component is an artifact. The first was the `find_bads_EOG_MNE` function which finds the most correlated components with the EOG electrodes. The 2nd method was the ICLabel model ported from EEGLAB. This model has been trained on identifying artifacts in the ICA components of EEG signals. Unfortunately, both methods resulted in worse results when compared to not using ICA at all. It is mentioned in some papers that ICA decreases correlation Wang P et al. 2023 [8]

Channel Selection sLORETTA was used in the "PreMovNet" paper. The spatial information gained was used to select 21 channels that are most correlated with the movement of the hand. We used the same channel selection for the first dataset "WAY-EEG-GAL"

3.2.2 "Full body in unconstrained motion". Data Importing: The data is stored in the format "AB_UH_(Participant Number)-T(Trial Number)-(eeg or kin).mat". The data importing script iterates through all of the files and checks if there is a corresponding kin file for each EEG file since there are some trials where one kind of data is missing. I found Participant #1 files to be corrupted so I didn't use it at all. The script stores the files in .npy files for each subject. In this dataset the EEG files are larger in size mainly due to the sampling frequency being double of the previous one (1000hz).

Data Preprocessing: In this dataset there is a mismatch between the sampling frequency of the EEG and the kinematic data, the EEG being 1000hz and the kinematic being 30hz. We tried 2 approaches to work this out. The first was upsampling the kinematic data to 1000hz. The 2nd approach was to predict only one kinematic point at the end of the EEG segment. The tables and graphs are only for the first approach because for the 2nd one our GAN architecture couldn't be used.

We used the same preprocessing techniques as the first dataset, First Bandpass Filtering was performed, then Common Average Referencing, as well as ICA. But the ICA lowered the correlation score so it wasn't used. We tried predicting different points of the participant's leg, and the results

were similar so we used one point "RightUpperLeg" for simplicity.

4 Methods

I experimented with different window sizes and step-sizes for the EEG segments, and found that a 500ms EEG window with 250ms step-size (50% overlap) performed the best, so that's what I went with for all the models.

4.1 CNN and LSTM

Taken directly from the "PreMovNet" paper to verify their results. The proposed CNN-LSTM model consists of nine layers that include batch normalization layer, two convolution layers, two max-pooling layers, one dropout layer, one LSTM layer, and two dense layers. The batch-normalization layer is represented as B1. The convolutional layer C1 has a kernel size of 7 and 256 filters, while C2 layer has a kernel size of 5 and 128 filters. The zero padding is done for both C1 and C2 layers so that the size of input and output remains the same. Each convolution layer is followed by a ReLu activation unit. M1 and M2 are max-pooling layers with a window size of 5 and 3, respectively. The layer D1 is a dropout layer with a 0.25 dropout rate. The layer L1 is the LSTM layer with 128 cells, followed by the ReLU activation function. The last two layers, D1 and D2, are dense layers with 128 and 3 neurons, respectively. The last layer with three output neurons yields the predicted hand kinematics in the x-, y-, and z-directions.

We used MSELoss, Adam Optimizer with learning rate of $3e-4$, and weight decay of $5e-2$.

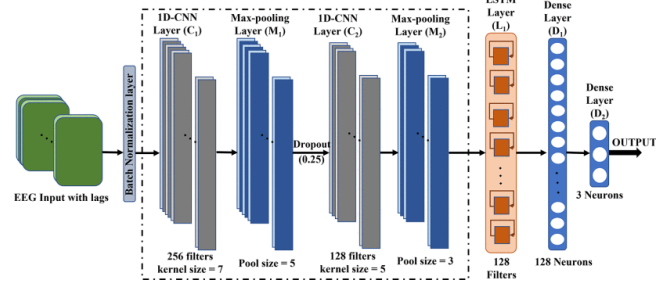


Figure 3. Architecture of the proposed PreMovNet-II model.

4.2 Modified Pix2Pix GAN

Pix2Pix GAN is a conditional GAN (cGAN) that was developed by Phillip Isola, Unlike vanilla GAN which uses only real data and noise to learn and generate images, cGAN uses real data and noise as well as labels to generate images. I found a PyTorch implementation of the model on Github. The original use of this GAN is to generate images. But, with the help of the signal-to-signal translation research by (Eric Easthope, 2024) [6], I modified the GAN for EEG signals by turning 2D convolutional layers, as well as the transpose

convolutional layers to 1D. I also squared the kernel size according to the research paper.

The Pix2Pix has two important architectures, one for the generator and the other for the discriminator, namely U-Net and PatchGAN.

U-Net generator: A symmetric encoder-decoder structure with down-sampling through max pooling (red arrows) and up-sampling via transposed convolutions (green arrows). Skip connections (gray arrows) connect layers of matching spatial dimensions in the encoder and decoder, preserving spatial information for segmentation in the output map.

PatchGAN discriminator: This architecture contains several transposed convolutional blocks. It takes an $N \times N$ part of the image/signal and tries to find whether it is real or fake. N can be of any size.

Loss Function: The goal of the generator $G(z)$ is to produce realistic fake samples, and the goal of the discriminator is to classify real data as real, and fake data as fake.

$D(x, y) = 1$ i.e. real and,

$D(x, G(z)) = 0$ i.e. fake.

The generator uses an L1 loss to produce signals close to the ground truth. As shown in equation (1)

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x, y, z} [\|y - G(x, z)\|_1]. \quad (1)$$

but more importantly it uses the GAN loss which uses the discriminator to learn the distribution of the data. instead of $D(x, G(z)) = 0$, we change $D(x, G(z)) = 1$.

$$\mathcal{L}_{cGAN}(G, D) = \mathbb{E}_{x, y} [\log D(x, y)] + \mathbb{E}_{x, z} [\log (1 - D(x, G(x, z)))] \quad (2)$$

And finally the Generator loss which adds them together

$$G^* = \arg \min_G \max_D \mathcal{L}_{cGAN}(G, D) + \lambda \mathcal{L}_{L1}(G). \quad (3)$$

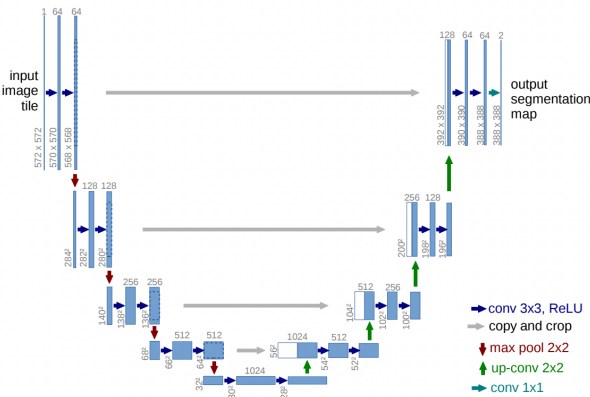


Figure 4. Architecture of the original U-Net.

4.3 Convolutional Autoencoder

Following a paper on transforming one biosignal to another (M. Haescher. et al, 2020) [7]. i.e Seismocardiograms (SCGs), into electrocardiograms (ECGs). We used the same architecture. "The autoencoder uses an encoder and a decoder with four one dimensional convolutional layers each. In the encoder, the convolutional layers are followed by a ReLU activation function for non-linearity, and a max-pooling layer for computational reduction. In the encoder, the first convolutional layer starts with 128 filters with a kernel size of 8. With each further layer, the number of filters is doubled. The latent space halves the number of filters. The decoder starts with 256 filters in the first convolutional layer. The second and third layer each half the number of filters. The final convolutional layer reduces the number of filters from 64 to 1. Each layer in our decoder consists of an upsampling layer, a convolutional layer, and a ReLU activation function".

5 Results

Time-series plots for the "WAY-EEG-GAL" dataset: The black lines are the ground truth data, and the red lines are the predicted data.

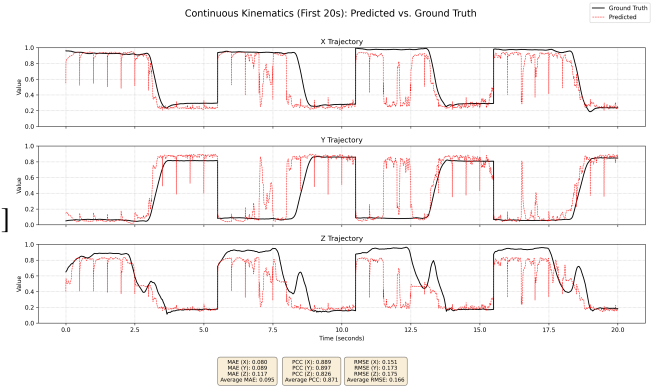


Figure 5. Position decoding using the CNN+LSTM model

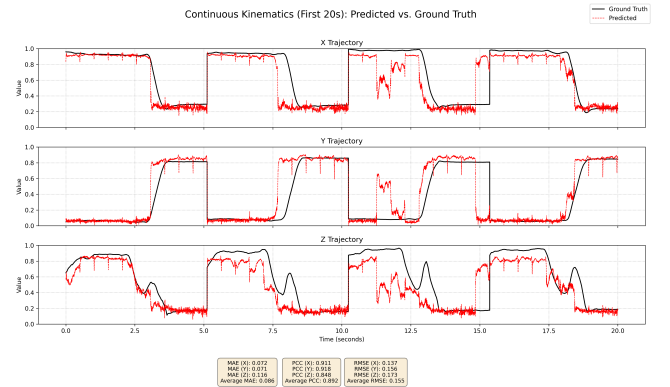


Figure 6. Position decoding using the GAN model

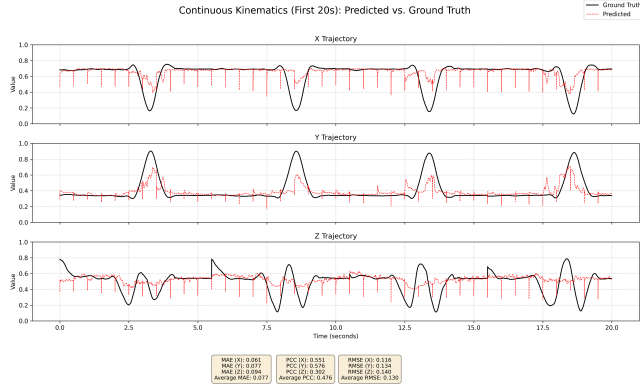


Figure 7. Velocity decoding using the CNN+LSTM model

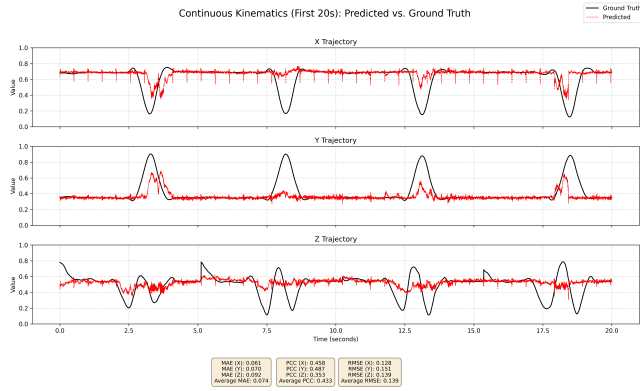


Figure 8. Velocity decoding using the GAN model

We can see from the position graphs that the CNN+LSTM model results in more sudden and steep to the signal whereas the GAN model is less erratic.

In the velocity graphs, the predicted data reacts similar to the real data in some of the peaks but in much smaller magnitude.

Time-series plots of the "Full body in unconstrained motion" dataset:

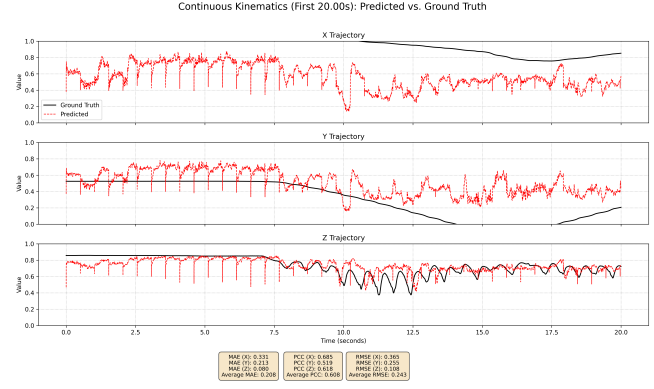


Figure 9. Position decoding using the CNN+LSTM model

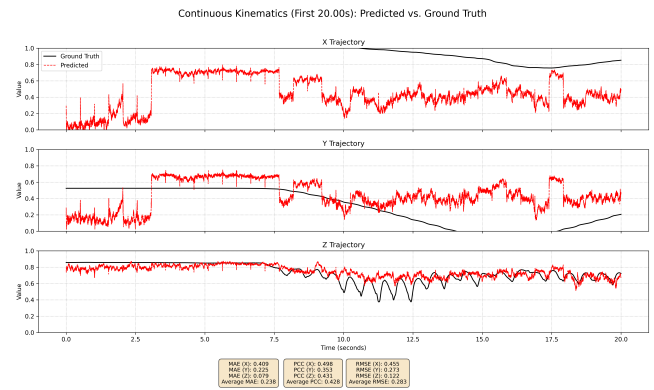


Figure 10. Position decoding using the GAN model

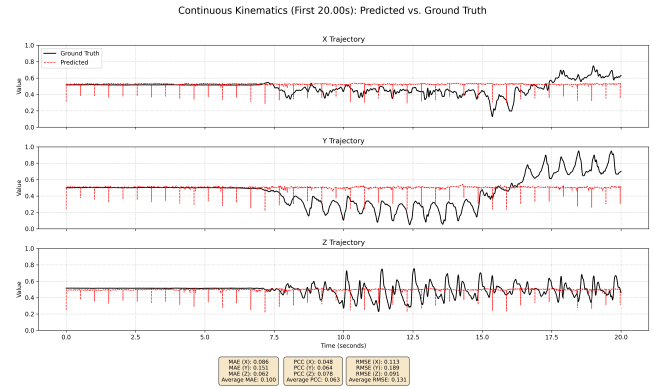


Figure 11. velocity decoding using the CNN+LSTM model

Table 1. Comparison of PCC scores between different time lags and models. Dataset "WAY-EEG-GAL"

Participant	Dir.	Lag								
		0 ms			150 ms			250 ms		
		CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE
P1	X	0.748	0.780	0.787	0.707	0.774	0.680	0.696	0.763	0.712
	Y	0.750	0.778	0.794	0.705	0.774	0.682	0.695	0.761	0.724
	Z	0.731	0.730	0.758	0.702	0.748	0.663	0.695	0.733	0.715
P2	X	0.684	0.680	0.594	0.694	0.633	0.662	0.654	0.642	0.527
	Y	0.695	0.688	0.591	0.701	0.639	0.675	0.662	0.650	0.538
	Z	0.674	0.656	0.601	0.694	0.616	0.648	0.673	0.643	0.566
P3	X	0.666	0.709	0.649	0.718	0.724	0.658	0.667	0.735	0.690
	Y	0.671	0.711	0.633	0.730	0.715	0.645	0.664	0.730	0.685
	Z	0.672	0.683	0.652	0.707	0.685	0.667	0.655	0.704	0.686
P4	X	0.806	0.814	0.765	0.816	0.838	0.761	0.820	0.842	0.770
	Y	0.815	0.824	0.789	0.825	0.850	0.795	0.821	0.849	0.803
	Z	0.751	0.792	0.766	0.760	0.802	0.734	0.765	0.818	0.771
P5	X	0.608	0.601	0.624	0.647	0.644	0.591	0.641	0.614	0.625
	Y	0.614	0.595	0.619	0.662	0.651	0.599	0.654	0.607	0.631
	Z	0.624	0.642	0.672	0.647	0.674	0.628	0.628	0.635	0.660
P6	X	0.615	0.669	0.602	0.605	0.669	0.562	0.595	0.657	0.499
	Y	0.599	0.653	0.591	0.588	0.652	0.566	0.577	0.635	0.498
	Z	0.556	0.614	0.543	0.563	0.608	0.527	0.548	0.597	0.454
P7	X	0.889	0.927	0.864	0.875	0.910	0.859	0.858	0.910	0.851
	Y	0.897	0.929	0.869	0.887	0.913	0.873	0.873	0.916	0.854
	Z	0.826	0.860	0.783	0.836	0.850	0.796	0.827	0.841	0.823
P8	X	0.434	0.431	0.301	0.487	0.424	0.365	0.450	0.449	0.433
	Y	0.450	0.449	0.291	0.494	0.437	0.342	0.446	0.458	0.384
	Z	0.458	0.491	0.345	0.494	0.476	0.402	0.454	0.472	0.451
P9	X	0.588	0.525	0.382	0.596	0.566	0.491	0.632	0.665	0.515
	Y	0.588	0.506	0.337	0.593	0.555	0.479	0.630	0.651	0.535
	Z	0.656	0.614	0.447	0.626	0.595	0.532	0.647	0.686	0.558
P10	X	0.876	0.867	0.809	0.842	0.850	0.729	0.815	0.842	0.799
	Y	0.880	0.872	0.813	0.852	0.844	0.746	0.823	0.849	0.807
	Z	0.847	0.819	0.775	0.836	0.810	0.693	0.814	0.824	0.779
P11	X	0.622	0.625	0.513	0.645	0.666	0.455	0.626	0.642	0.457
	Y	0.634	0.621	0.593	0.659	0.666	0.466	0.641	0.626	0.465
	Z	0.663	0.666	0.445	0.693	0.701	0.488	0.681	0.673	0.520
P12	X	0.850	0.804	0.779	0.837	0.847	0.789	0.841	0.842	0.792
	Y	0.848	0.809	0.785	0.840	0.846	0.792	0.845	0.846	0.815
	Z	0.817	0.811	0.779	0.827	0.837	0.814	0.827	0.845	0.819
Average	X	0.699	0.703	0.639	0.706	0.712	0.634	0.691	0.717	0.639
	Y	0.703	0.703	0.642	0.711	0.712	0.638	0.694	0.715	0.645
	Z	0.690	0.698	0.631	0.699	0.700	0.633	0.685	0.706	0.650

Table 2. Comparison of PCC scores between different frequency bands and models with no time lag. Dataset "WAY-EEG-GAL".

Participant	Dir.	Frequency Band								
		0.5–3 Hz			0.5–12 Hz			0.1–40 Hz		
		CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE
P1	X	0.448	0.566	0.436	0.506	0.575	0.501	0.748	0.780	0.787
	Y	0.448	0.553	0.439	0.504	0.563	0.491	0.750	0.778	0.794
	Z	0.391	0.529	0.419	0.464	0.536	0.486	0.731	0.730	0.758
P2	X	0.356	0.261	0.172	0.466	0.512	0.402	0.684	0.680	0.594
	Y	0.361	0.252	0.176	0.470	0.523	0.404	0.695	0.688	0.591
	Z	0.348	0.279	0.158	0.487	0.533	0.441	0.674	0.656	0.601
P3	X	0.353	0.520	0.367	0.555	0.631	0.530	0.666	0.709	0.649
	Y	0.360	0.525	0.369	0.566	0.635	0.540	0.671	0.711	0.633
	Z	0.362	0.496	0.408	0.535	0.618	0.546	0.672	0.683	0.652
P4	X	0.551	0.481	0.418	0.559	0.633	0.522	0.806	0.814	0.765
	Y	0.566	0.494	0.423	0.569	0.642	0.533	0.815	0.824	0.789
	Z	0.464	0.418	0.337	0.421	0.562	0.463	0.751	0.792	0.766
P5	X	0.420	0.421	0.290	0.519	0.606	0.429	0.608	0.601	0.624
	Y	0.418	0.415	0.266	0.521	0.604	0.425	0.614	0.595	0.619
	Z	0.449	0.441	0.317	0.524	0.622	0.451	0.624	0.642	0.672
P6	X	0.343	0.460	0.387	0.426	0.472	0.374	0.615	0.669	0.602
	Y	0.348	0.477	0.390	0.424	0.485	0.393	0.599	0.653	0.591
	Z	0.318	0.428	0.342	0.395	0.438	0.367	0.556	0.614	0.543
P7	X	0.498	0.556	0.461	0.642	0.781	0.644	0.889	0.927	0.864
	Y	0.493	0.558	0.456	0.649	0.780	0.646	0.897	0.929	0.869
	Z	0.449	0.515	0.410	0.596	0.722	0.593	0.826	0.860	0.783
P8	X	0.310	0.323	0.276	0.448	0.495	0.391	0.434	0.431	0.301
	Y	0.322	0.325	0.279	0.452	0.490	0.385	0.450	0.449	0.291
	Z	0.312	0.334	0.298	0.461	0.495	0.414	0.458	0.491	0.345
P9	X	0.067	0.235	0.145	0.046	0.277	0.163	0.588	0.525	0.382
	Y	0.085	0.222	0.148	0.041	0.273	0.177	0.588	0.506	0.337
	Z	0.062	0.195	0.139	0.061	0.261	0.200	0.656	0.614	0.447
P10	X	0.504	0.602	0.494	0.490	0.596	0.485	0.876	0.867	0.809
	Y	0.514	0.604	0.501	0.501	0.603	0.497	0.880	0.872	0.813
	Z	0.458	0.555	0.440	0.449	0.570	0.459	0.847	0.819	0.775
P11	X	0.472	0.396	0.260	0.487	0.402	0.378	0.622	0.625	0.513
	Y	0.482	0.395	0.265	0.502	0.401	0.377	0.634	0.621	0.593
	Z	0.478	0.394	0.281	0.487	0.408	0.322	0.663	0.666	0.445
P12	X	0.641	0.669	0.537	0.760	0.751	0.665	0.850	0.804	0.779
	Y	0.622	0.661	0.540	0.756	0.756	0.653	0.848	0.809	0.785
	Z	0.585	0.611	0.519	0.704	0.733	0.624	0.817	0.811	0.779
Average	X	0.414	0.458	0.354	0.492	0.561	0.457	0.699	0.703	0.639
	Y	0.418	0.457	0.354	0.496	0.563	0.460	0.703	0.703	0.642
	Z	0.390	0.433	0.339	0.465	0.542	0.447	0.690	0.698	0.631

Table 3. Comparison of RMSE scores between different frequency bands and models. Dataset "WAY-EEG-GAL".

Participant	Dir.	Frequency Band								
		0.5–3 Hz			0.5–12 Hz			0.1–40 Hz		
		CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE
P1	X	0.343	0.354	0.383	0.332	0.348	0.358	0.253	0.244	0.236
	Y	0.385	0.386	0.425	0.373	0.382	0.403	0.301	0.279	0.270
	Z	0.289	0.285	0.301	0.277	0.283	0.301	0.211	0.213	0.203
P2	X	0.320	0.396	0.411	0.303	0.328	0.347	0.251	0.280	0.302
	Y	0.352	0.421	0.434	0.332	0.348	0.375	0.269	0.298	0.325
	Z	0.303	0.347	0.368	0.280	0.293	0.306	0.234	0.275	0.267
P3	X	0.350	0.348	0.385	0.316	0.312	0.331	0.279	0.268	0.305
	Y	0.388	0.378	0.422	0.343	0.337	0.361	0.307	0.293	0.340
	Z	0.325	0.322	0.335	0.292	0.288	0.301	0.256	0.268	0.278
P4	X	0.254	0.307	0.313	0.253	0.264	0.282	0.181	0.180	0.205
	Y	0.290	0.352	0.366	0.289	0.303	0.327	0.209	0.206	0.233
	Z	0.283	0.328	0.350	0.288	0.288	0.324	0.206	0.191	0.206
P5	X	0.252	0.304	0.327	0.238	0.256	0.294	0.223	0.262	0.240
	Y	0.265	0.317	0.346	0.263	0.285	0.315	0.255	0.282	0.263
	Z	0.283	0.313	0.326	0.266	0.259	0.297	0.242	0.259	0.234
P6	X	0.267	0.297	0.302	0.258	0.293	0.310	0.239	0.254	0.266
	Y	0.293	0.308	0.318	0.275	0.302	0.324	0.274	0.282	0.274
	Z	0.274	0.286	0.295	0.265	0.282	0.293	0.241	0.254	0.277
P7	X	0.283	0.288	0.312	0.253	0.205	0.260	0.151	0.127	0.164
	Y	0.347	0.359	0.388	0.297	0.254	0.315	0.173	0.146	0.196
	Z	0.275	0.274	0.304	0.254	0.218	0.256	0.175	0.167	0.195
P8	X	0.397	0.444	0.454	0.373	0.390	0.427	0.379	0.436	0.451
	Y	0.396	0.438	0.446	0.373	0.385	0.425	0.375	0.427	0.443
	Z	0.321	0.346	0.356	0.300	0.311	0.329	0.301	0.334	0.359
P9	X	0.341	0.399	0.399	0.340	0.392	0.410	0.283	0.330	0.372
	Y	0.378	0.448	0.441	0.380	0.437	0.454	0.318	0.370	0.411
	Z	0.308	0.365	0.342	0.307	0.350	0.355	0.237	0.271	0.311
P10	X	0.295	0.297	0.322	0.297	0.298	0.326	0.163	0.173	0.206
	Y	0.313	0.312	0.337	0.316	0.310	0.342	0.174	0.186	0.216
	Z	0.349	0.336	0.362	0.351	0.330	0.362	0.212	0.225	0.244
P11	X	0.282	0.342	0.365	0.279	0.331	0.320	0.249	0.283	0.313
	Y	0.308	0.373	0.396	0.304	0.366	0.349	0.280	0.322	0.302
	Z	0.271	0.297	0.319	0.266	0.286	0.283	0.224	0.240	0.266
P12	X	0.237	0.250	0.291	0.199	0.219	0.249	0.162	0.189	0.204
	Y	0.296	0.295	0.336	0.246	0.254	0.297	0.199	0.224	0.237
	Z	0.232	0.236	0.257	0.201	0.200	0.229	0.163	0.173	0.179
Average	X	0.310	0.344	0.365	0.295	0.319	0.325	0.232	0.248	0.264
	Y	0.343	0.374	0.392	0.324	0.340	0.361	0.251	0.276	0.293
	Z	0.293	0.307	0.322	0.282	0.286	0.295	0.215	0.227	0.238

Table 4. Comparison of PCC scores between different frequency bands and models with no time lag. Dataset "Full body in unconstrained locomotion".

Participant	Dir.	Frequency Band								
		0.5–3 Hz			0.5–12 Hz			0.1–40 Hz		
		CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE
P1	X	0.227	0.266	0.206	0.263	0.193	0.241	0.235	0.263	0.225
	Y	0.196	0.174	0.212	0.312	0.243	0.274	0.240	0.224	0.235
	Z	0.472	0.413	0.567	0.594	0.417	0.510	0.537	0.483	0.585
P2	X	0.267	0.339	0.245	0.251	0.273	0.251	0.260	0.385	0.358
	Y	0.196	0.204	0.203	0.202	0.185	0.195	0.242	0.360	0.302
	Z	0.396	0.578	0.432	0.454	0.527	0.416	0.385	0.435	0.490
P3	X	0.307	0.313	0.310	0.381	0.479	0.279	0.474	0.437	0.475
	Y	0.275	0.251	0.212	0.292	0.379	0.227	0.358	0.347	0.408
	Z	0.596	0.458	0.495	0.639	0.435	0.533	0.500	0.388	0.340
P4	X	0.511	0.394	0.400	0.554	0.424	0.377	0.685	0.498	0.435
	Y	0.434	0.380	0.317	0.474	0.345	0.346	0.519	0.353	0.396
	Z	0.545	0.100	0.142	0.580	0.153	0.218	0.618	0.431	0.407
P5	X	0.283	0.298	0.285	0.277	0.331	0.292	0.227	0.263	0.293
	Y	0.232	0.202	0.207	0.245	0.249	0.216	0.268	0.225	0.296
	Z	0.404	0.506	0.215	0.356	0.331	0.289	0.335	0.227	0.100
P6	X	0.287	0.249	0.245	0.300	0.292	0.242	0.259	0.383	0.223
	Y	0.388	0.386	0.366	0.418	0.383	0.339	0.469	0.509	0.440
	Z	0.343	0.354	0.251	0.368	0.422	0.246	0.353	0.409	0.294
P7	X	0.229	0.278	0.166	0.322	0.248	0.199	0.292	0.216	0.161
	Y	0.291	0.261	0.260	0.363	0.247	0.281	0.365	0.322	0.226
	Z	0.386	0.339	0.315	0.463	0.460	0.304	0.390	0.359	0.239
P8	X	0.288	0.328	0.189	0.288	0.380	0.259	0.425	0.472	0.459
	Y	0.361	0.394	0.287	0.389	0.421	0.313	0.573	0.541	0.400
	Z	0.363	0.205	0.335	0.467	0.277	0.258	0.614	0.578	0.483
P9	X	0.031	0.092	0.025	0.087	0.087	-0.005	0.152	0.114	0.133
	Y	0.049	0.054	-0.013	0.214	0.101	-0.011	0.143	0.140	0.142
	Z	0.085	0.038	0.069	0.205	0.071	0.117	0.149	0.109	0.117
Average	X	0.270	0.284	0.230	0.303	0.301	0.237	0.334	0.337	0.307
	Y	0.269	0.256	0.228	0.323	0.284	0.242	0.353	0.336	0.316
	Z	0.399	0.332	0.313	0.458	0.344	0.321	0.431	0.380	0.339

Table 5. Comparison of RMSE scores between different frequency bands and models with no time lag. Dataset "Full body in unconstrained locomotion".

Participant	Dir.	Frequency Band								
		0.5–3 Hz			0.5–12 Hz			0.1–40 Hz		
		CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE	CNN+LSTM	GAN	CAE
P1	X	0.195	0.231	0.224	0.200	0.249	0.210	0.201	0.218	0.216
	Y	0.223	0.271	0.241	0.205	0.243	0.225	0.213	0.229	0.236
	Z	0.092	0.098	0.089	0.084	0.097	0.091	0.090	0.092	0.085
P2	X	0.204	0.223	0.217	0.207	0.216	0.212	0.211	0.206	0.201
	Y	0.189	0.222	0.193	0.187	0.195	0.191	0.188	0.189	0.187
	Z	0.100	0.090	0.098	0.097	0.093	0.101	0.101	0.098	0.095
P3	X	0.149	0.179	0.151	0.145	0.138	0.158	0.147	0.152	0.143
	Y	0.243	0.284	0.257	0.243	0.240	0.271	0.235	0.271	0.243
	Z	0.077	0.089	0.084	0.075	0.089	0.081	0.083	0.089	0.092
P4	X	0.369	0.454	0.404	0.408	0.400	0.396	0.365	0.455	0.348
	Y	0.288	0.291	0.311	0.256	0.292	0.287	0.255	0.273	0.317
	Z	0.113	0.153	0.149	0.108	0.143	0.140	0.108	0.122	0.126
P5	X	0.284	0.295	0.292	0.288	0.309	0.290	0.314	0.333	0.307
	Y	0.301	0.324	0.314	0.301	0.354	0.311	0.313	0.349	0.318
	Z	0.074	0.070	0.081	0.078	0.078	0.077	0.078	0.081	0.086
P6	X	0.248	0.286	0.258	0.244	0.253	0.259	0.260	0.267	0.282
	Y	0.212	0.230	0.231	0.209	0.229	0.230	0.202	0.211	0.224
	Z	0.251	0.269	0.267	0.251	0.251	0.266	0.256	0.259	0.274
P7	X	0.279	0.313	0.328	0.277	0.330	0.319	0.273	0.335	0.326
	Y	0.178	0.186	0.196	0.179	0.211	0.190	0.173	0.198	0.208
	Z	0.184	0.199	0.204	0.178	0.182	0.204	0.184	0.205	0.205
P8	X	0.272	0.274	0.323	0.269	0.264	0.291	0.258	0.260	0.270
	Y	0.252	0.255	0.286	0.251	0.250	0.268	0.221	0.237	0.270
	Z	0.203	0.241	0.226	0.192	0.240	0.233	0.176	0.188	0.205
P9	X	0.312	0.366	0.313	0.311	0.354	0.313	0.314	0.389	0.369
	Y	0.312	0.394	0.317	0.303	0.376	0.317	0.332	0.386	0.374
	Z	0.208	0.262	0.232	0.204	0.260	0.236	0.217	0.264	0.243
Average	X	0.257	0.292	0.279	0.267	0.286	0.277	0.260	0.306	0.285
	Y	0.244	0.276	0.262	0.237	0.275	0.257	0.237	0.273	0.266
	Z	0.145	0.164	0.155	0.141	0.160	0.151	0.144	0.163	0.157

6 Discussion

6.1 "WAY-EEG-GAL"

The GAN results have the highest correlation but the difference is small between the models. The highest correlation was achieved in the 250ms time lag. There is a stark difference between some of the participants such as P7, and P8 from an almost perfect 0.9 correlation to a 0.4 mediocre correlation. This suggests that some artifact removal is needed since I choose to opt out of ICA. ICA done with expert help would increase the correlation scores. Reasons our results do not match with the previously mentioned paper are that the paper downsampled both the EEG and the Kinematic data but we kept it the same i.e 500Hz. Meaning our model is predicting 5 times the data so it is more prone to errors. Another reason is lack of clarity on Windows and Steps sizes, and whether their model predicted a complete kinematic sequence or a kinematic point at the end of the EEG sequence. When we tried the prediction of a single point, the results were improved.

6.2 "Full body in unconstrained motion"

The results are quite worse than the previous dataset. Even though the sampling frequency and the number of electrodes is double, the decoding is hard due to the unpredictable nature of the executed movements. The EEG setup is now mobile and the movement events are complex with movement of the whole body instead of just one limb. There's been previous studies of decoding lower limb movement but they involved the participant moving on a treadmill which means the movement is very controlled and repetitive, which is easier to decode. In this dataset, all of the complex movements introduce many artifacts to the data. To improve on the model, we can use biomechanical models of the lower limb to capture the intrinsic joint angles performed. Or we can just add a constraint (distance between joints) that punishes the model with loss when it is exceeded. A similar method was utilized in (P. Wang. et al, 2023) [3].

6.3 Velocity

The PCC scores for the velocities of the wrist with **0.1-40hz** EEG features were quite low with an average of **0.3612, 0.3648, 0.1985** for the X,Y,Z directions respectively. An interesting improvement happened when I decreased the frequency range to **0.5-3Hz**. The PCC score reached an average of **0.4771, 0.4732, 0.3038** for X,Y,Z. After that, I tried the decoding but with a time lag of 250ms, and the score improved again to an average of **0.5204, 0.5343, 0.2813**. Decreasing the frequency range to 0.5-3Hz and adding a time lag to the EEG signal has improved results in decoding velocity components. On the contrary the same adjustments to the frequency and time lag have made the results worse for the decoding of position components.

6.4 Conclusion

In conclusion. Decoding motion data from EEG signals is possible but with many limitations. The results are good when the EEG recording is in highly controlled environments with clearly split trials of repetitive movements, which is the case in the first dataset. However, in the 2nd dataset there are more varied and complex movements with rest in between. In future work, it's worth focusing on the following points

- Optimizing and improving the velocity decoding performance of the model since velocity should be more intuitive.

- Implement more expert artifact removal with ICA to fix the drastic difference of performance between some of the participants in the 1st dataset.

- Add geometric constraints to the model to punish for impossible joint angles, and incorrect limb lengths.

- Identify the rest periods in the 2nd dataset's trials and delete them so the data left is only movement.

- Select only the important electrodes which correlate the most with the movement data to prevent confusing the model with too much data.

Supplementary Methods

The code, processed datasets, and graphs are available through the following links:

- **Code repository:** https://github.com/yoitshussam/EEG_Motion_Trajectory_Reconstruction
- **Graphs:** <https://drive.google.com/drive/folders/1ToxyUmJrijwxiuPIfit>
- **Processed Datasets:** <https://drive.google.com/drive/folders/14ycG6U5O6Tdyd7cylbwa9F6Qcsls6xsP?usp=sharing>

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